

Agile Methodology

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Introduction

Agile Methodology is a modern software development approach that focuses on flexibility, collaboration, and continuous improvement. Instead of developing an entire application at once, Agile divides the project into smaller development cycles called **iterations** or **sprints**. At the end of each sprint, a working product increment is delivered to the customer for review and feedback.

Agile encourages frequent communication among team members, customers, and stakeholders. It helps organizations quickly adapt to changing requirements while maintaining high software quality and customer satisfaction.

What is Agile Methodology?

Agile is an iterative and incremental software development methodology where development activities are divided into small, manageable tasks. Each sprint usually lasts between one and four weeks, during which planning, development, testing, and review are completed.

The primary objective of Agile is to deliver working software quickly while responding effectively to customer feedback and changing business needs.

Agile Manifesto

The Agile Manifesto defines four core values:

1. **Individuals and interactions** over processes and tools.
2. **Working software** over comprehensive documentation.
3. **Customer collaboration** over contract negotiation.
4. **Responding to change** over following a fixed plan.

These values encourage continuous collaboration and adaptability throughout the software development lifecycle.

Principles of Agile

The Agile methodology is based on several important principles:

- Deliver working software frequently.
- Welcome changing requirements, even during development.
- Maintain close collaboration between business users and developers.

- Build projects around motivated individuals.
- Ensure continuous attention to technical excellence.
- Keep the design simple.
- Regularly reflect and improve team performance.

Agile Lifecycle

The Agile software development lifecycle consists of the following phases:

1. Requirement Gathering

Customer requirements are collected and prioritized based on business value.

2. Planning

The project is divided into multiple sprints, and tasks are assigned to the development team.

3. Design

System architecture and design documents are prepared for the selected sprint.

4. Development

Developers implement the planned features during the sprint.

5. Testing

The developed features are tested continuously to ensure software quality.

6. Deployment

Completed features are deployed for customer evaluation.

7. Review and Feedback

Customers review the product increment and provide feedback, which is incorporated into future sprints.

Agile Scrum Framework

Scrum is one of the most widely used Agile frameworks.

Scrum Roles

- Product Owner
- Scrum Master
- Development Team

Scrum Events

- Sprint Planning
- Daily Scrum
- Sprint Review

- Sprint Retrospective

Scrum Artifacts

- Product Backlog
- Sprint Backlog
- Product Increment

Advantages of Agile Methodology

- Faster software delivery
- Improved customer satisfaction
- Better communication among team members
- Higher product quality
- Continuous testing and integration
- Early identification of defects
- Flexibility to accommodate changing requirements
- Reduced project risk
- Increased team productivity
- Faster time-to-market

Limitations of Agile

Although Agile provides many benefits, it also has some challenges:

- Requires active customer participation.
- Documentation may be limited.
- Difficult to estimate project cost for large projects.
- Requires experienced and collaborative team members.
- Scope changes may affect project timelines.

Comparison between Agile and Waterfall

Agile	Waterfall
Iterative development	Sequential development
Flexible requirements	Fixed requirements
Continuous customer involvement	Limited customer interaction
Frequent software delivery	Final delivery after project completion

Easy to accommodate changes	Changes are expensive
Continuous testing	Testing after development

Popular Agile Frameworks

Several frameworks implement Agile practices:

- Scrum
- Kanban
- Extreme Programming (XP)
- Lean Software Development
- Crystal
- Feature Driven Development (FDD)
- Dynamic Systems Development Method (DSDM)

Applications of Agile

Agile methodology is widely used in:

- Software Development
- Web Application Development
- Mobile Application Development
- Cloud Computing Projects
- Artificial Intelligence Projects
- DevOps Projects
- Data Science Applications
- Product Development
- Game Development

Benefits for Organizations

Organizations adopting Agile experience:

- Better customer engagement
- Improved product quality
- Faster release cycles
- Continuous innovation
- Improved project visibility

- Reduced development cost
- Better team collaboration
- Higher return on investment

Conclusion

Agile Methodology has transformed software development by promoting flexibility, collaboration, and continuous improvement. By dividing projects into short development cycles and encouraging customer feedback, Agile enables organizations to deliver high-quality software quickly and efficiently. Its iterative approach helps teams respond to changing business requirements while maintaining customer satisfaction and product quality. Agile has become one of the most widely adopted software development methodologies across industries because of its adaptability and effectiveness.